Appendix A: Transcript of initial interaction with the client

Date: 19 September, 2024

Location: Mr. Weatherby's study

Client: Mr. Weatherby

Interviewer: Student Developer

Me: Good afternoon, Mr. Weatherby, and thank you for taking the time to meet with me today. My name is Ryan and I'm currently working on developing a dining hall routine software for teachers, suggested by Mrs. Stewart and Mr. Pitt, a part of my career science internal assessment. After a discussion with them, I think you're the best person to be my client in the school, as you are in charge of everything about routine. The purpose of this meeting is to gather some requirements and insight from you so that I can design a solution that truly meets your needs effectively. To start, I would like to understand how the current system operates. Can you walk me through the existing process of managing dining hall routines?

Client: Yes, okay. So every day, every day, Monday through to Saturday, there are five supervision slots. Breakfast, early lunch, which is 12.30 to 1 o'clock, 1 o'clock until 2 o'clock pretty much, and that counts as two slots. We need two members of staff, and then dinner duty. So that's how we, and we need members of staff there for each of those slots every day of the week, Monday through to Saturday. So at the moment, the assistant house parent writes to all the tutors in their boarding house and says, can you let me know when you're not free? And then the assistant house parent places everybody else into slots throughout the week to fill all the gaps. So every tutor normally does about four or five duties a week. There are some slots which are already filled because staff who are part of what we call the pastoral or academic leadership team, they have a weekly fixed slot. So, for

example, I take on one week, on one of the weeks, I take a Thursday slot, and on one of the weeks, I take a Tuesday slot. So there's less slots than what it seems, but there's still a good number of slots that need filling. So if someone's free, if someone can't make a Thursday lunch, but everything else is free, they're normally allocated maybe a breakfast, two lunch times and a dinner slot, that sort of thing. And that fills out the week.

Me: Okay, that sounds great. And from your experience, what do you think the system, in what way does it, what is the advantage of it? What do you think?

Client: Well, the advantage is, it's a difficult one. We've thought long and hard about how we do the staffing of the dining hall slots. It's good that the assistant house parent does it at the moment because they obviously know their tutor team. Staff can also, to a certain extent, pick which slots suit them best or which slots they can't make. So there should be no clashes with the teacher's teaching timetable because they've already identified which slots they can't do. So everything else, in theory, they are free to do. So there should be no clashes there already.

Me: Okay. But what sort of challenge have you, or limitation have you seen with this system?

Client: It is challenging because some slots are more popular than other slots. So not everybody wants to do a breakfast supervision because if you don't live close to the school, then it's quite difficult for you to get in here to start supervising at 7 o'clock in the morning. Similarly, evening dinner time is not a popular supervision. It tends to be the night that you are doing your duty night in-house. You might say, well, if I have to stay on anyway to do that duty, I'll also roll it into a dining hall supervision. The Saturday dinner slot is not a popular slot and often no one volunteers for that slot. So I do that slot. So there are challenges. There is occasionally a slot where no

teachers of that house are actually able to make. Everybody's teaching. So sometimes there's a slot which we can't fill with teachers of that boarding house. So we have to ask if anybody else for that week would be free to take that slot. And very occasionally we might say to the head boy and head girl, can you supervise that slot?

Me: Do you think there are any really fixed rules that the piece of software must follow?

Client: Well, I suppose you need to fill all the slots. It's a given. You need to fill all the slots. You'd probably say that no member of staff should do more than four or five slots. And that everybody who's a tutor, if it's your house doing the next week, every tutor probably should do no less than, say, three or four slots. Because you want a reasonably fair distribution of work. And that's difficult to get without. In reality, there'll be some tutors who do maybe a little bit more than others. So now let's go on to the software component. So in terms of functionality, what do you think are some essential features that there should be in the software? Well, I don't know. I don't know exactly how the software will work. I've never done it electronically. I've always done it manually. I dare say in a perfect world, if you are able somehow to record which slots members of staff can't do, and then if the system could allocate everybody to fill the slots which are available and share out the workload, then that would be a perfect scenario. So I don't know how that would work. You'd have to have some system where you said, let's just say I'm a tutor. And I said, I can't do Tuesday early lunch, and I can't do Thursday dinner, and I can't do Friday breakfast. Then every other slot is available to me. And if you've got seven tutors, six or seven tutors, all doing the same thing, and then the system can hopefully allocate everybody fairly to all the gaps and avoid allocating you to a slot that you've already indicated you can't do.

Me: Are there any additional or desirable features you would like to include?

Client: No, I don't know. I've never really thought about using software before, so I don't know whether there would be a desirable feature. Obviously you want it to be as easy as you possibly can. If you allocated each person a letter, A, B, C, D, etc., and then in each cell you recorded whether, let's just say I couldn't make Tuesday lunchtime, so somehow that Tuesday lunchtime slot, and if I'm a letter A, somehow you had it so that slot, you'd have to go through and pre-populate which cells certain people couldn't do. You'd have to say not A for that cell. Then I don't know how you can randomly generate people's names across there, but test each time it was going to allocate a person's name to a cell to test whether it was not A for that cell if I'm letter A. I don't know how you could do it. I'm not experienced in writing programs.

Me: What would you say about the minimum requirement you want the software to do?

Client: It needs to use all the tutor staff for that boarding house, and it needs to try and allocate a fair workload to all tutors so that one person doesn't get 10 slots and another person gets one slot. That's the minimum.

Me: Okay. Do you think there are any additional rules that are out of this software or guidelines I should be aware of when I will be designing this software?
Client: Not really. Not that I can think of. In theory, there should be enough staff to cover all the slots, and I guess if everyone says they can't make a certain slot, then that slot will be left blank, and then it would require a manual process to find a solution for who can fill that blank slot.

Me: Okay. Do you think I will get the access to the data I need, for example, like a timetable of a certain teacher?

Client: Well, you wouldn't need a timetable. I think you would need to work with one of the assistant house parents as they are about to perform this manual allocation because they will write initially and say to the members of staff, tell me which slots you can't make. They could then share that with you. You could operate a blank grid of which slots need filling, and then you could see maybe the assistant house parent could do what they usually do and fill it manually and you could use your software to fill it electronically and then compare. Yeah, I think that's what we'll do at the start of when I will be doing it. But I think you should be working with your assistant house parent, and maybe you can either go back and work with Kimmy, or I can give you a list of which houses are about to do which weeks. Okay. Yeah.

Me: Okay. Are there any other things you want to add?

Client: Do you know, I'd be very hopeful if this can work. Yeah. It might make life a lot easier for our assistant house parents because I know it's not a task that they relish doing. Every house does a dining hall supervision probably four times a year. So each time it comes around, it's not a fun job, but it's a necessary job, so we need it done.

Yeah. Okay. I think that sounds great. Thank you again for your time and your insight. Your feedback will be valuable for me to design an efficient software. I'll use this information to move forward with my development, and I'll keep you updated with my progress.

Client: Terrific. Okay. Thank you.